Poll Everywhere Quizzes

At the beginning of each lecture:

1. Use your device to log in to PollEverywhere (PE). Enter your ANU email (uXXXXXXX@anu.edu.au); PE supports ANU SSO.

   https://pollev.com/anuiml2026



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Probability spaces and random variables

Lecture 2 - Quiz 1 (120 seconds)

L2Q1 - In a large population of credit applicants, 30% have a low income, 25% have high outstanding debt, and 20% have a short employment history. An applicant is drawn at random. What can be concluded about the probability that they have **at least one** of the three risk factors?

- A) It equals $0.30 + 0.25 + 0.20 = 0.75$.
- B) It is at least 0.75, since the probability of a union is never smaller than the sum of the individual probabilities.
- C) It is at most 0.75, and the exact value cannot be determined without knowing how the three groups overlap.
- D) Nothing can be concluded, because the union bound only applies when the three groups are disjoint.

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Lecture 2 - Quiz 2 (120 seconds)

L2Q2 - A server's response time, in hours, is modelled as Uniform[0, 1/5]. A student computes $f_X(0.1) = \frac{1}{1/5-0} = 5$ and objects: *"a probability of* 5 *is absurd, so the model must be wrong"*. What is the correct response?

- A) The value 5 is correct, but it is a density, not a probability: for example $\mathbb{P}(X = 0.1) = 0$, and probabilities are areas, so $\mathbb{P}(0.1 \leq X \leq 0.15) = 0.05 \times 5 = 0.25$.
- B) The density has been computed incorrectly: for a Uniform[0, 1/5] it should equal 1/5 rather than 5, and with the correct value no absurdity arises.
- C) The value 5 is correct, but a density must be divided by the total area underneath it before it can be read as the probability of the corresponding value.
- D) The error is modelling a response time as a continuous quantity: it should be described by a probability mass function, and a mass function can never exceed 1.

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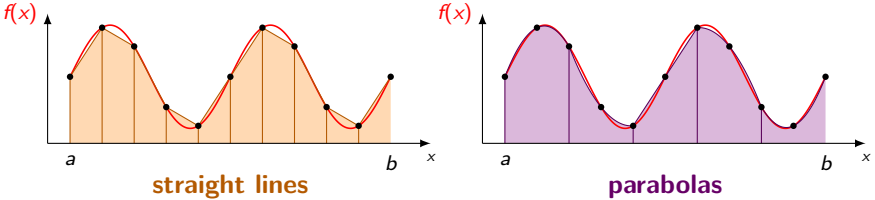
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Expectation, the integral and variance

Lecture 2 - Quiz 3 (120 seconds)

L2Q3 - We must approximate the area under f between a and b, using the **same** 10 subintervals in both rules. Which rule should we prefer?

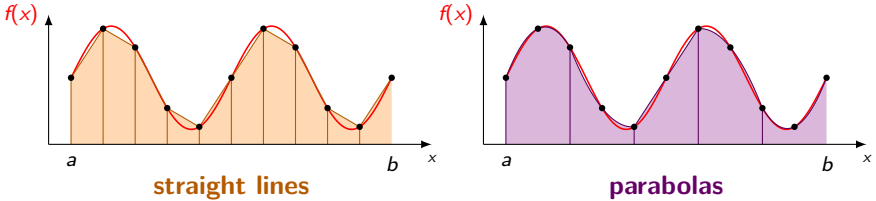


- A) The trapezoidal rule, because straight lines never overshoot the curve, whereas parabolas can.
- B) Simpson's rule, because on this curve the straight lines cut visibly across every peak and trough, while the parabolas follow the curve almost exactly.
- C) Simpson's rule, because for the same number of subintervals it evaluates f at twice as many points as the trapezoidal rule.
- D) Neither: both rules use the same 11 points, so they must give the same approximation.

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Lecture 2 - Quiz 4 (120 seconds)

L2Q4 - Two delivery services are compared. Service A always delivers in exactly 3 days. Service B delivers in 1 day or in 5 days, each with probability $\frac{1}{2}$. A manager notes that both have an expected delivery time of 3 days and concludes that the two services are equivalent. What is the best assessment?

- A) The manager is correct: the expectation is the average outcome, so two random variables with the same expectation behave the same on average, and that is all one can ask of them.
- B) The manager is wrong, because Service B's expected delivery time is not 3: an expectation must be a value the random variable can actually take, and B never delivers in exactly 3 days.
- C) The manager is correct as long as the two services have the same variance, and that cannot be decided from the information given.
- D) The manager is wrong: both expectations really are 3, but the expectation says nothing about spread. Service B has a far larger variance, and its expected value is not even a possible delivery time.

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Conditioning, target function and sampling

Lecture 2 - Quiz 5 (120 seconds)

L2Q5 - Features take three values, each with probability $\frac{1}{3}$, and $\eta(x) = \mathbb{P}(Y = 1 \mid \mathbf{X} = x)$ satisfies

$$\eta(x_1) = 0.1, \qquad \eta(x_2) = 0.6, \qquad \eta(x_3) = 0.9$$

Recall that under the binary loss the error of a rule at a given x is the probability that its prediction differs from Y at that x, and the out-of-sample error is the average of these errors over the distribution of $\mathbf{X}$. What is the target function f^* and its out-of-sample error?

- A) f^* outputs (0.1, 0.6, 0.9), since the target function is the conditional expectation η.
- B) f^* outputs $(0, 1, 1)$ and $E_{\text{out}}(f^*) = 0$, since f^* is optimal.
- C) f^* outputs $(0, 1, 1)$ and $E_{\text{out}}(f^*) = \frac{1}{3}(0.1 + 0.4 + 0.1) = 0.2$.
- D) f^* outputs $(0, 1, 1)$ and $E_{\text{out}}(f^*) = \frac{1}{3}$, since it is wrong on one of the three feature values.

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Lecture 2 - Quiz 6 (120 seconds)

L2Q6 - A hospital predicts a disease from blood pressure. It has 1000 measurements, but they come from only 100 patients, 10 consecutive daily readings each, and the team treats this as a sample with $N = 1000$. What is the problem?

- A) The observations are not independent, since readings on the same patient are largely predictable from one another, so the effective sample size is far below 1000 and the i.i.d. guarantees do not apply.
- B) There is no problem: 1000 measurements are 1000 observations, and the i.i.d. assumption concerns only how the patients were selected, not how many readings each of them contributes.
- C) The observations are not identically distributed, since blood pressure drifts from one day to the next, but they remain independent, so the sample can be used exactly as it stands.
- D) The problem is the size of the sample: $N = 1000$ is simply too small for any generalisation guarantee to be useful, however the measurements happen to be collected.

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Lecture 2 - Quiz 7 (120 seconds)

L2Q7 - In Lecture 1 we assumed an **ideal rule** f mapping each input to its output. In Lecture 2 we saw that two applicants with **identical** recorded features may repay or default. How are these two statements reconciled?

- A) The Lecture 1 picture was simply wrong: once the same input can lead to different outcomes there is no ideal rule to be found, and the supervised framework has to be rebuilt without any notion of a target function.
- B) The randomness is only measurement noise: with precise enough instruments and enough recorded features the outcome would be completely determined by the input, so the ideal rule f of Lecture 1 is deterministic after all.
- C) The target function maps each $\mathbf{x}$ to the outcome actually observed for it in the data: since the dataset is finite each $\mathbf{x}$ occurs once, and the conflict only appears if we imagine drawing further applicants with the same features.
- D) The target function is defined through the conditional distribution of Y given $\mathbf{X}$: under the binary loss it predicts the most probable label at each $\mathbf{x}$. It is the *best* rule, not a rule that is always right, and in general its error is greater than zero.

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Lecture 2 - Quiz 8 (120 seconds)

L2Q8 - Two teams model the same phenomenon on the same 200-example dataset. Team A uses $\mathcal{H}_A =$ **all** functions from $\mathcal{X}$ to $\{0, 1\}$; Team B uses the perceptron. Both report $E_{\text{in}}(\hat{h}) = 0$. Whose zero is more informative?

- A) Team A, because a larger hypothesis set contains rules closer to the target function, so its approximation error is smaller and its fit is therefore more trustworthy.
- B) Team B: $\mathcal{H}_A$ can fit any dataset whatsoever, so a zero $E_{\text{in}}(\hat{h})$ there is unfalsifiable and carries no information, whereas the same fit from a restricted hypothesis set is informative.
- C) Neither: both teams achieved $E_{\text{in}}(\hat{h}) = 0$, and $E_{\text{in}}(\hat{h})$ is the only quantity we can actually compute, so from the data alone the two situations are indistinguishable.
- D) Team A, because obtaining $E_{\text{in}}(\hat{h}) = 0$ from a much larger hypothesis set is a stronger achievement than obtaining it from a small one such as the perceptron.

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